AUTHOR TESTED VIDEO PROOF PROVIDED

A Futuristic Wearable HEALTH MONITORING RING

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While working on an IoT based data acquisition system using health monitoring sensors, we started visualising this health monitoring system, which could be worn like an ornament in our day to day life. We thought, what if it's a smart device that can store real-time data, creating a platform for both monitoring and evaluation for a doctor while we are in our home itself. It could have an SOS feature, which is one of the main issues in India where people are unable to contact hospitals or doctors well in time.

Let us try and build the system that can take care of the above-mentioned problems. A private channel can be created for the person whose health parameters are to be monitored. There can be provision for an SOS emergency indication to alert the doctor or any other designated person. Author's prototype of the system on a breadboard is shown in Fig. 1.

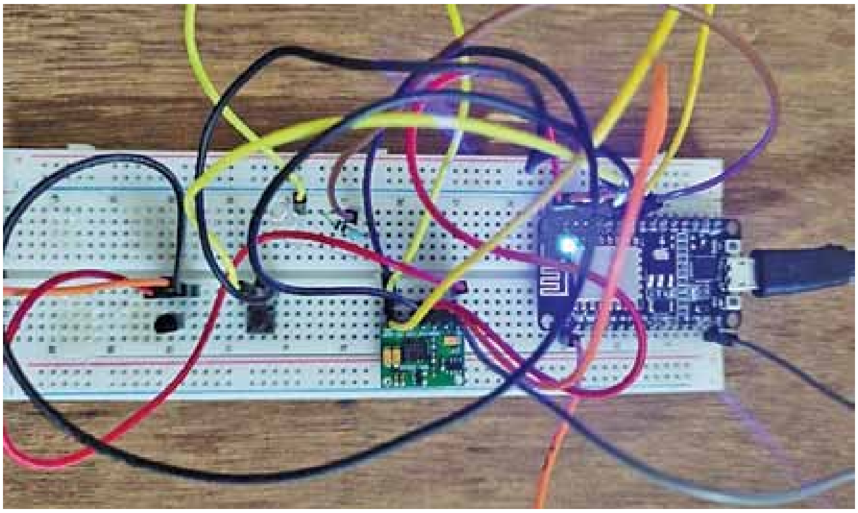

Fig. 1: Author's prototype

The idea is to enclose the whole system in a small-size ornament like a ring to monitor the health parameters. So, we can replace the components in the prototype with small-size components that can fit in a ring whose design is of our choice. The prototype can be designed in the shape of a dice shaped ring, as shown in Fig. 2 and Fig. 3, or any other practical shape of our choice.

Circuit and working

Circuit diagram of the smart wearable health monitoring ring built around nodeMCU ESP8266 microcontroller is shown in Fig. 4. The other components include MAX30100 pulse oximeter sensor, temperature sensor LM35, a small pushbutton switch for triggering an SOS message, an LED for on/off indication, and a tiny 3.3V battery.

BILL OF MATERIAL

Component name	Quantity	Description
NodeMCU ESP8266	1	Microcontroller to correlate the entire process and connect to cloud
MAX 30100 pulse oximeter sensor	1	To measures pulse and spo2 level
Push button	1	Switch
LM35	1	Temperature measuring device
LED and resistor	1 each	Indication of action
Jumper wires	As required	Used for connection
Lithium-ion battery	5V/2000mAh	Used to power the device

ThingSpeak is used to create the private cloud based IoT-enabled monitoring platform with data retrieval capability. (ThingSpeak is an IoT analytics platform service that allows you to aggregate, see, and analyse live data streams in the cloud.) You read the data from the public ThingSpeak Channel 12397 Weather Station and write it into your new channel.

You can sign into ThingSpeak using your MathWorks account credentials, or create a new account. After that click on Channels >

Fig. 2: Ring design